

TASK -1

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix  . :  
IPv6 Address. . . . . : 2401:4900:ab43:e9a4:7441:fcde:4391:a38  
Temporary IPv6 Address. . . . . : 2401:4900:ab43:e9a4:d4e4:ab17:e2b:d1c2  
Link-local IPv6 Address . . . . . : fe80::1072:d1af:6458:471b%6  
IPv4 Address. . . . . : 172.20.10.3  
Subnet Mask . . . . . : 255.255.255.240  
Default Gateway . . . . . : fe80::7414:d0ff:fe59:8264%6  
                             172.20.10.1
```

Ethernet adapter Ethernet:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix  . :
```

TASK-2

C:\Users\raftaar>ping www.google.com

```
Pinging www.google.com [2001:4860:4826:7700::] with 32 bytes of data:  
Reply from 2001:4860:4826:7700::: time=98ms  
Reply from 2001:4860:4826:7700::: time=110ms  
Reply from 2001:4860:4826:7700::: time=131ms  
Reply from 2001:4860:4826:7700::: time=141ms
```

```
Ping statistics for 2001:4860:4826:7700::  
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:  
Minimum = 98ms, Maximum = 141ms, Average = 120ms
```

TASK-3

C:\Users\raftaar>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [2600:140f:7200:49::17d4:70]
over a maximum of 30 hops:

```
 1      2 ms      2 ms      2 ms  2401:4900:ab43:e9a4:f4:7913:f433:35bc  
 2      *          *          *      Request timed out.  
 3    133 ms    101 ms    101 ms  fc00:0:2:1f3::1  
 4    114 ms    12 ms     14 ms  2404:a800:2a00:100::6  
 5    135 ms    101 ms    101 ms  2404:a800:2a00:100::5  
 6    120 ms    100 ms    101 ms  2404:a800::42  
 7    189 ms    104 ms    100 ms  ae4.r02.border01.bom01.fab.netarch.akamai.com [2600:140f:7600:ffff::252]  
 8    110 ms    105 ms    100 ms  ae2.r02.stem01.bom01.fab.netarch.akamai.com [2600:140f:7600:303::1]  
 9    115 ms    104 ms    102 ms  ae1.r01.spine02.bom01.fab.netarch.akamai.com [2600:140f:7600:c01::]  
10    116 ms    100 ms    101 ms  ae1.r02.leaf01.bom01.fab.netarch.akamai.com [2600:140f:7600:c02::1]  
11    111 ms    101 ms    100 ms  ae2.r13.tor01.bom01.fab.netarch.akamai.com [2600:140f:7600:50d::1]  
12    205 ms    103 ms    102 ms  g2600-140f-7200-0049-0000-0000-17d4-0070.deploy.static.akamaitechnologies.com [2600:140f:7200:49::17d4:70]
```

Trace complete.

C:\Users\raftaar>|

TASK-4

```
C:\Users\raftaar>nslookup www.spotify.com
Server: UnKnown
Address: fe80::7414:d0ff:fe59:8264

Non-authoritative answer:
Name: atc.spotify.map.fastly.net
Addresses: 2a04:4e42:31::810
           151.101.211.42
Aliases: www.spotify.com

C:\Users\raftaar>
```

TASK-5

Recommended Troubleshooting Sequence

1. Check Local IP Configuration: ipconfig (or ifconfig / ip a on Mac/Linux)

- **What it checks:** Verifies whether your friend's laptop is properly connected to the local Wi-Fi/Ethernet network and has been assigned a valid **IPv4 address** and **Default Gateway** (router IP).
- **Diagnosis:** If the IPv4 address starts with 169.254.x.x (Auto-IP) or shows blank, the laptop isn't getting an IP address from the router, pointing to a local Wi-Fi or DHCP issue.

2. Test DNS Domain Resolution: nslookup www.zomato.com

- **What it checks:** Determines if the DNS server can successfully translate the domain name www.zomato.com into its actual IP address.
- **Diagnosis:** If it returns an error like DNS request timed out or Server failed, the issue is a **DNS failure** (which can be fixed by changing DNS settings to Google DNS 8.8.8.8 or Cloudflare 1.1.1.1). If it returns a valid IP address, DNS is working fine.

3. Test Connectivity & Routing Path: ping www.zomato.com (or tracert www.zomato.com)

- **What it checks:** Tests end-to-end packet reachability and measures latency between the laptop and Zomato's servers.

- **Diagnosis:**

- If ping returns "**Request timed out**" or 100% packet loss, the server is unreachable or blocked.
- Running tracert will trace the exact hop where data stops moving, showing if the connection breaks at the local ISP, an intermediate WAN router, or Zomato's edge firewall.